Diabetes 130-US Hospitals for Years 1999-2008

Donated on 5/2/2014

The dataset represents ten years (1999-2008) of clinical care at 130 US hospitals and integrated delivery networks. Each row concerns hospital records of patients diagnosed with diabetes, who underwent...

Dataset Characteristics
Multivariate

Subject Area
Health and Medicine

Associated Tasks
Classification, Clustering

Feature Type
Categorical, Integer

Instances
101766

Features
47

Citations/Acknowledgement

If you use this dataset, please cite:

Please cite:

Beata Strack, Jonathan P. DeShazo, Chris Gennings, Juan L. Olmo, Sebastian Ventura, Krzysztof J. Cios, and John N. Clore, “Impact ...

Creators

John Clore

Krzysztof Cios

Dataset Information

What do the instances in this dataset represent?

The instances represent hospitalized patient records diagnosed with diabetes.

Are there recommended data splits?

No recommendation. The standard train-test split could be used. Can use three-way holdout split (i.e., train-validation-test) when doing model selection.

Does the dataset contain data that might be considered sensitive in any way?

Yes. The dataset contains information about the age, gender, and race of the patients.

Additional Information

The dataset represents ten years (1999-2008) of clinical care at 130 US hospitals and integrated delivery networks. It includes over 50 features representing patient and hospital outcomes. Information was extracted from the database for encounters that satisfied the following criteria....

SHOW MORE

Has Missing Values?

Yes

Introductory Paper

[Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Record]

By Beata Strack, Jonathan DeShazo, Chris Gennings, Juan Olmo, Sebastian Ventura, Krzysztof Cios, John Clore. 2014
Published in BioMed Research International, vol. 2014

Variables Table

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
repaglinide	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up		no

Jon DeShazo

Beata Strack

DOI

10.24432/C5230J

License

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		
nateglinide	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
chlorpropamide	Feature	Categorical		The feature indicates whether the drug was prescribed or		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		
glimepiride	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
acetohexamide	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
glipizide	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				no if the drug was not prescribed		
glyburide	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
tolbutamide	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				decreased, steady if the dosage did not change, and no if the drug was not prescribed		
pioglitazone	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
rosiglitazone	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		
acarbose	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
miglitol	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		
troglitazone	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
tolazamide	Feature	Categorical		The feature indicates whether the drug was		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		
examide	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
citoglipton	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
insulin	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				no if the drug was not prescribed		
glyburide-metformin	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
glipizide-metformin	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				decreased, steady if the dosage did not change, and no if the drug was not prescribed		
glimepiride-pioglitazone	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
metformin-rosiglitazone	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		
metformin-pioglitazone	Feature	Categorical		The feature indicates whether the drug was prescribed or there was a change in the dosage. Values: up if the dosage was increased during the encounter, down if the dosage was decreased, steady if the dosage did not change, and no if the drug was not prescribed		no
change	Feature	Categorical		Indicates if there was a change in diabetic medications (either dosage or generic name).		no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
				Values: change and no change		
diabetesMed	Feature	Categorical		Indicates if there was any diabetic medication prescribed. Values: yes and no		no
readmitted	Target	Categorical		Days to inpatient readmission. Values: <30 if the patient was readmitted in less than 30 days, >30 if the patient was readmitted in more than 30 days, and No for no record of readmission.		no

Rows per page 25 25 to 50 of 50

Additional Variable Information

Detailed description of all the atrributes is provided in Table 1 Beata Strack, Jonathan P. DeShazo, Chris Gennings, Juan L. Olmo, Sebastian Ventura, Krzysztof J. Cios, and John N. Clore, “Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records,” ...

SHOW MORE

Dataset Files

File	Size
diabetic_data.csv	18.3 MB
IDS_mapping.csv	2.5 KB

Reviews

3.8 (3 ratings)

LOGIN TO WRITE A REVIEW

Leland Burrill

3/30/2025

I fit an xgboost model w/ 0.96 accuracy and 0.99 AUC. I currently work with this kind of data. The trick here is to realize some patients have multiple records, and to engineer features that leverage that history, i.e, their past admissions. A histo...

SHOW MORE

Michael Pluda

1/4/2025

Not a bad dataset, but definitely not for beginners or intermediates. This is a very imbalanced dataset, and even if you manage to soften this issue with under/oversampling techniques, the chance of distinguishing between the...

SHOW MORE

Vincent Mischa

11/16/2024

how to solve the missing value of weight when data preprocessing ? cause there're way too much.